The Commit

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COVER STORY · MACBOOK PRO

Apple brings touch and OLED to MacBook Pro

The new professional laptops combine a different display architecture with touch controls, with US prices starting at $2,899.

IMAGE: APPLE

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Apple's new 14- and 16-inch laptops arrive November 6. Their stacked OLED layers share the work of producing light, while a display engine compensates for changes in individual pixels.

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git log --oneline

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01 COVER STORY MacBook Pro

Apple brings touch and OLED to MacBook Pro

The new professional laptops combine a different display architecture with touch controls, with US prices starting at $2,899.

src: apple.com · Apple · 26 min read

Apple has introduced 14- and 16-inch MacBook Pro models with a tandem OLED display and the first touch capabilities on a Mac. Preorders opened October 28, with availability beginning November 6. US prices start at $2,899 for the 14-inch M5 Pro model and $3,399 for its 16-inch counterpart. M5 Max configurations are also available.

The display's two light-emitting layers share the work, allowing each to operate at lower current. Apple says this improves brightness, efficiency and lifespan. Its supplied technical paper specifies 1,000 nits of sustained full-screen brightness in environments up to 25°C, alongside peaks of 2,000 nits. Sustained and peak output are different measures.

A new display engine continuously compensates for usage, temperature and changing pixel performance. Apple describes this as burn-in mitigation, rather than a guarantee against ageing. Touch also brings direct interaction to creative applications, including playing onscreen instruments in Logic Pro. For buyers doing colour-sensitive work, the calibration and uniformity claims will matter alongside the new controls.

02 TECH RADAR

DHS employee records exposed

A leak exposed details of over 41,000 Homeland Security employees and contractors, WIRED reports, including sensitive undercover-program information. Okta denies evidence its systems were compromised.

src: wired.com

Driving patterns trigger police stops

The ACLU warns police are using licence-plate records and AI to flag supposedly suspicious travel patterns, extending surveillance beyond investigations of specific crimes.

src: gizmodo.com

Stanford authors challenge book piracy

Stanford authors allege Anthropic downloaded and stored pirated books. The Stanford Daily reports a court distinction between unlawful acquisition and fair-use training.

src: stanforddaily.com

Washington challenges Utah over Kalshi

The US commodities regulator sued Utah to block enforcement against prediction markets. Utah calls Kalshi an illegal casino; federal officials assert regulatory preemption.

src: arstechnica.com

03 AI AI security

Google restricts its new cyber-capable Gemini

Gemini 4 Carbon goes first to selected cyber defenders while Google tests its safeguards before a wider release.

src: blog.google · 2026-10-28

Google has announced Gemini 4 Carbon, which it calls its strongest model, but most people cannot use it yet. Starting this week, access goes to a subset of cyber defenders, with more trusted partners to follow through the company's Fairwind Program.

Google links the cautious rollout to the model's security abilities. It reports state-of-the-art scores of 88.2% for finding vulnerabilities on SEC-bench Pro, 54.8% for exploitation on ExploitGym and 71.0% for patching on CWE-bench. Carbon also ranks first on version 5.0 of the Artificial Analysis Intelligence Index, at 59.

The company is taking part in the US government's voluntary process for testing models before release. It says it will keep gathering feedback and adjusting guardrails before offering Carbon to developers, businesses and consumers.

Google also says Carbon is built for long engineering tasks: its teams have seen it work for days on subtle bugs across large codebases. In internal testing, the company reports that Carbon roughly halves unsupported claims compared with Argon.

These are Google's own measurements. Until access widens, outside users cannot check whether the results hold up in everyday security work, and Google has given no date for general availability.

04 MODEL WATCH

Tue 27 → Thu 29 Oct

NEW FROM THE LABS	SIZE	CAPABILITIES
Gemini 4 Carbon GOOGLE · 28 OCT	– undisclosed	
Genie 4 GOOGLE DEEPMIND · 27 OCT	– undisclosed	
Nemotron 3.5 Transcribe NVIDIA · 28 OCT	**30B** 4B active	
TRENDING ON HUGGING FACE	**SIZE**	**CAPABILITIES**
DeepSeek V4.1 Flash DEEPSEEK · #1	**763.2B**	
LFM3-1.7B LIQUID AI · #6	**1.73B**	
MiniMax M3.1 Flash MINIMAX · #7	**428B** 23B active	

Reasoning · Vision · Coding · Agents & tool use · Speech & audio · Video · Open weights

06 SEMICONDUCTORS Tue 27 → Thu 29 Oct

OPTICAL INTERCONNECTS

Imec maps the next steps beyond copper

Bringing light closer to processors promises bandwidth gains, but materials and manufacturing still constrain the roadmap.

src: semiengineering.com · 2026-10-27

Imec maps how light-based links could move closer to AI chips, from front panels to chip packages and wafer stacks. Its report says copper links get harder to scale as data rates and distance rise. They draw more power, add heat and lose signal quality. These limits can hold back AI systems as data needs grow.

Plug-in optical modules already turn electrical signals into light sent over fibre. Silicon photonics has helped them reach 200 gigabits per second per lane. For future 3.2-terabit modules, imec points to new materials that could help turn electrical signals into light with less loss.

This is a roadmap, with no claim of mass output at those future speeds. Micro-transfer printing could place new materials onto silicon, but work to scale it is still at the research stage. Chip packages must also cope with heat and meet cost and reliability needs before these plans can yield practical systems.

07 CHIP BRIEFING

Prototype memory stores overlapping AI vectors

IBM Research Europe researchers demonstrated phase-change memory that stores multiple AI vectors in superposition, increasing effective storage density at a tunable accuracy cost. Tests on prototype hardware achieved inference accuracy close to full-precision software, according to the research summary. This is an experimental memory architecture, with no commercial production claim supplied.

src: semiengineering.com

Power semiconductors could shrink grid transformers

Solid-state transformers combine magnetic components with high-power semiconductors, offering smaller devices and greater control of electricity flows, MIT Technology Review reports. They could help address grid pressure from data centres and other demand. The technology remains in development and cannot yet replace every conventional transformer, especially the largest high-voltage transmission units.

src: technologyreview.com

Chip models must reveal manufacturing consequences

Advanced packaging makes suppliers' manufacturing choices affect neighbouring components. Semiconductor Engineering examines how models can share temperature response, warping and process variation without exposing proprietary recipes. Fixed material values can conceal real operating risks, a Brewer Science researcher warns. The issue is predictive modelling across companies, rather than a new production milestone.

src: semiengineering.com

08 EXPLAINER full piece ~3 min

BROWSER FINGERPRINTING

A suspicious script needs more than a name

Browser fingerprinting uses rendering differences to distinguish devices. A reverse-engineering account shows why alarming code labels require careful interpretation.

src: nullpt.rs · nullpt.rs · 2026-10-28

A researcher investigating browser fingerprinting found an unusually obscured script and published an account with a provocative question about bioweapons. The supplied text supports a narrower explanation: this was an investigation of browser identification and misleading code. It does not establish that a website created a biological weapon, or that an alarming name describes what software actually does.

Canvas fingerprinting begins with an ordinary browser feature: drawing text, shapes or other graphics onto a canvas. The script then extracts the resulting pixels. Browsers, operating systems, graphics drivers and hardware can render the same instructions slightly differently. Those differences can help distinguish one browser environment from another. The useful signal is in the rendered result, rather than simply the words or shapes being drawn.

To find examples, the author crawled sites from the Tranco Top Sites list using Playwright. The crawler intercepted calls that extract canvas images and checked properties associated with fingerprinting or bot detection, including navigator.webdriver. The author collected examples at fingerprint.garden, finding familiar fingerprinting products as well as the script that prompted this investigation.

The account identifies a script hosted at servicepipe.tech and says its variable names appeared intended to mislead language-model-assisted analysis. The author asked Astra to examine it with that possibility in mind. The text also describes apparently malicious code gated behind an environment flag when running under Node. The supplied excerpt does not establish that this branch executed in visitors' browsers.

The author's overall assessment is that, beyond its unusual obfuscation, the script appears to be a standard anti-bot script. That leaves a useful distinction for readers: evidence of fingerprinting, interpretation of misleading names, and proof of harmful execution are separate questions. This account gives an explanation of the first and reasons to investigate the others carefully.

BYTE-SIZED JOKE

My backup drive joined a gym. It wanted stronger recovery options.

09 COMPANY ACTIONS Tue 27 → Thu 29 Oct

Search and cloud lift Alphabet revenue

Alphabet reported quarterly revenue of $127.6 billion, up 25% from a year earlier. Google Cloud revenue rose 103% to $30.7 billion, while Search and other revenue grew 12%. Chief executive Sundar Pichai attributed faster search-query growth to AI investment. The figures show expansion across those businesses, while his explanation of what drove searches remains the company's account of its performance.

ALPHABET · theverge.com

Azure grows as consumer businesses contract

Microsoft reported revenue of $92.3 billion and net income of $33.6 billion for its first fiscal quarter of 2027. Azure revenue rose 48% to $33.17 billion as revised reporting disclosed the cloud business separately. Devices and Consumer revenue fell 3%, with Windows OEM and devices down 22%. The results underline how cloud growth now outweighs weakness in consumer hardware businesses.

MICROSOFT · theverge.com

Record profit concentrates Samsung's memory exposure

Samsung reported quarterly operating profit of 107.44 trillion won, broadly matching analyst estimates, CNBC reports. Its semiconductor business contributed 106.5 trillion won, showing how heavily profits depend on chips. Samsung cited increased HBM4 sales and server storage demand, and plans more advanced-memory sales and long-term contracts. Its expectation of tighter supply in 2027 remains a company forecast rather than an established outcome.

SAMSUNG · cnbc.com

ServiceNow expands its OpenAI integration plans

OpenAI and ServiceNow announced broader integration of models and agents into business workflows. The ServiceNow MCP plugin is available now, while enhanced capabilities are expected by year-end. Planned security features enter early access in January 2027, with general availability targeted for the second quarter. Existing permissions will govern workflow actions, OpenAI says; much of the expansion remains scheduled rather than delivered.

OPENAI / SERVICENOW · openai.com

Google builds Singapore health research ties

Google announced a Singapore research hub and a Ministry of Health partnership intended to expand healthcare AI. Google.org and DeepMind are contributing $7 million to a regional health coalition, including disease-detection work. Google says a clinical information assistant is being evaluated with the National University Health System. Evaluation, research funding and deployment ambitions do not themselves establish improved patient outcomes.

GOOGLE · blog.google

M6 iMac starts at $1,499

Apple announced an M6 iMac starting at $1,499 in the US, with availability beginning November 6. The entry configuration has a 10-core CPU and 10-core GPU; a 12-core version starts at $1,699. Apple claims faster productivity, graphics and local AI performance against earlier models. Those task-specific comparisons are company measurements, useful to distinguish from a universal speed improvement across every application.

APPLE · apple.com

10 COMPANY ACTIONS CONTINUED

Tue 27 → Thu 29 Oct

Echo Studio adds an eye to Alexa

Amazon's Echo Studio with Home Camera adds a 13-megapixel camera to its smart speaker for $250, or $30 above the camera-free version. Amazon says the camera can help maintain perceived volume as listeners move, check a room remotely and let Alexa recognise objects. The core speaker drivers and processor match the standard model, Ars Technica reports. Amazon says a dedicated button disables microphones and cameras, and an LED indicates cloud image or video streaming. Buyers therefore face a choice about visual sensing and privacy alongside audio features. Fire TV video calling is promised without a launch date.

A AMAZON · arstechnica.com

Vulcan targets November return after booster problems

United Launch Alliance plans to return Vulcan to flight on November 13, pending range approval, carrying 40 Amazon Leo broadband satellites. The rocket has been grounded since a February booster-nozzle anomaly. Nozzle failures affected two of its first four missions, although both flights completed their objectives. ULA says design and process changes mitigate potential contributors, but neither it nor booster maker Northrop Grumman has publicly identified the latest failure's root cause. Additional testing will continue alongside the planned flight. The mission matters for Amazon's stored satellites and the US military launch backlog; successful return remains to be demonstrated.

U ULA · arstechnica.com

11 MARKETS

27 Oct → 28 Oct

S&P 500	NASDAQ	EURO STOXX 50	BITCOIN
▲ +0.3%	▲ +0.3%	▲ +0.1%	▲ 0.0%
7,744	26,550	6,168	80,805

▲ BIGGEST GAINERS

+10.7%	Oracle ORCL	$145
+4.1%	Arm ARM	$236
+3.2%	AMD AMD	$578

▼ BIGGEST LOSERS

−1.2%	Meta META	$612
−1.2%	Nvidia NVDA	$215
−1.0%	Palantir PLTR	$197

Change between daily closes · finance.yahoo.com · not investment advice

12 WORLD REPORT Tue 27 → Thu 29 Oct

Spain investigates deadly train dispatch error

A Spanish rail collision killed at least 68 people on Wednesday, France 24 reports. The transport minister said a train had been ordered to proceed in error despite a technical failure that halted services. The initial findings concern Castilla-La Mancha and renew safety questions after another high-speed collision killed 46 people in January.

SPAIN · france24.com

Arab-majority parties poised for historic gains

Two Arab-majority parties were projected to win a combined 19 seats after Tuesday's Israeli election, France 24 reports. The result remained provisional as counting continued. The report describes record turnout and anger among Arab voters at Benjamin Netanyahu's government. These voters belong to a community representing around a fifth of Israel's population.

ISRAEL · france24.com

Canada and EU shelve associate-membership wording

Canada and the European Union dropped proposed associate-membership language from a draft summit declaration seen by Reuters, France 24 reports. They instead seek closer association, with further work continuing into next year. Diplomats said Germany led objections to an undefined legal concept. Prime Minister Mark Carney wants to reduce Canada's economic and security dependence on the US.

CANADA / EU · france24.com

Ebola infections pass 10,000

More than 10,000 people have been infected in the Ebola outbreak declared in May, according to Congolese government figures reported by NPR. Nearly 5,000 people have died, the report says. The outbreak spans Nigeria to Kenya. Delayed detection of the Bundibugyo species and conflict in northeastern Congo complicated the response; the epidemic remains ongoing.

CENTRAL AFRICA · npr.org

Kherson aid ceasefire expands, officials say

A ceasefire intended to deliver aid to Oleshky has expanded across the wider Kherson region, Ukrainian officials and Melania Trump said. The BBC reports that both sides said the truce was being observed. Humanitarian deliveries are to be followed by voluntary evacuations planned for Friday. Thousands remain trapped in Oleshky without essential supplies and services.

UKRAINE · bbc.co.uk

Flávio Bolsonaro arrives in Miami

Brazilian Senator Flávio Bolsonaro arrived in Miami after losing Sunday's presidential runoff, the BBC reports. The US provided visas after Brazil cancelled his passports during a criminal investigation. He denies wrongdoing and says he did not flee. Lula da Silva's reelection was confirmed by the top electoral court; Bolsonaro's departure adds a diplomatic dimension to the dispute.

BRAZIL · bbc.co.uk